

Manuel Mezo

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[Github](#) | [Project portfolio](#) | [LinkedIn](#)

PROFILE

Aerospace and Mechanical Engineer turned Robotics and Computer Vision builder, blending hands-on expertise in **ROS2**, custom **ML models**, and physical **hardware deployment** with an **8+ year** track record of solving multi-million-dollar operational challenges at **Amazon** and **McKinsey**. Now independently developing and deploying functional **robotic systems** and **AI vision applications**, bringing a rare combination of hardware/software engineering capability and large-scale operational rigor to the AI and Robotics industry.

TECHNICAL SKILLS

Robotics & Simulation: ROS2, Gazebo, CoppeliaSim, **Arduino**, Inverse Kinematics, 3D Printing, CAD Design; **Computer Vision & ML:** OpenCV, YOLO (fine-tuning, ~97% mAP@50), Ultralytics, Roboflow, ByteTrack; **Development:** Python, C++, SQL, Git/GitHub, CI/CD; **Cloud & Data:** Google Cloud, AWS, Pandas, NumPy, Tableau, Power BI

COMPUTER VISION AND ROBOTICS PROJECTS

- **Custom YOLOv8 Object Tracking & Spatial Logic:** Engineered a custom detection pipeline using an annotated dataset in Roboflow. Fine-tuned a **YOLOv8n model** (~97% mAP@50) and implemented real-time tracking via **ByteTrack** and **OpenCV**. Developed advanced spatial logic for moving conveyor belts featuring point-in-polygon zones.
- **Integrated VLA AI model in 3D Printed Robotic Arm:** Built a physical 4-axis robotic arm integrated with a cloud-based **Vision-Language-Action (VLA)** reasoning engine. The system utilizes **Gemini 1.5 Pro** to translate natural language voice commands into precise mechanical actuations.
- **Self-Balancing Robot & ROS2 Navigation:** Developed a physical robot on **Arduino** hardware, implementing IMU filtering and **PID control loops**. Built a **ROS2** simulation in **Gazebo** to parse **LIDAR** data for autonomous navigation.
- **VisionAI & Real-Time Traffic Analyzer:** Developed a browser-based dashboard for real-time video telemetry. Built **VisionAI**, an application powered by **Gemini 2.5 Flash** that detects and tracks objects via natural language queries.

APPLIED AI PROJECTS

- **AI Voice Customer Service Agent:** Deployed voice-first agent (**Retell.ai + Voiceflow**) with real-time order lookups via Vercel serverless functions.
- **X-Ray Medical Diagnostic Agent:** Autonomous computer vision agent built on **Gemini 2.5 Flash** for real-time medical image analysis.
- **WMS with Camera Vision:** Mobile-first inventory management system using on-device camera for barcode scanning and **Firebase** synchronization.
- **SupplyChain-Swarm-Sim:** Multi-agent adversarial simulation using **Claude Code swarms** to stress-test logistics network resilience.

PROFESSIONAL EXPERIENCE

Manager, EU Transportation Forecasting | Amazon (Madrid, Spain)

Feb 2026 – Present

- Lead the technical roadmap for EU transportation forecasting, targeting **>\$10M/year** in savings through **ML bias correction**, automated simulation modules, and data pipeline health checks.
- Own end-to-end production of weekly demand forecasts covering all EU logistics network segments, driving transportation and labor capacity optimization.
- Manage a distributed team of **6 Business Intelligence Engineers** and Program Managers; lead roadmap prioritization with the Forecasting Tech engineering team.
- Drive continuous improvement of forecast generation infrastructure, overseeing model retraining cycles, anomaly detection, and automated alerting.

Sr Program Manager, Supply Chain Planning | Amazon (Madrid, Spain)

Sept 2023 – Feb 2026

- Invented and implemented a **long-range (52-week) forecasting methodology** for EU truckload demand, supporting S&OP cycles and driving **\$140Mn CapEx decisions** for **2025**.
- Extended the forecasting horizon for inbound flows to **12 weeks**, improving planning for **800k annual labor hours**.
- Improved **inventory transfer truck fill rate modeling** (inaccuracy reduced from **35%** to **17%**), saving **~\$1.8Mn annually** by preventing late additions and cancellations.
- Identified and resolved a critical logic defect in the automated scheduling system using **SQL-based data analysis** and **Python scripting**, protecting **~\$10M** in avoidable late truck addition costs.

Operations Consultant | McKinsey & Company (Madrid, Spain)

Aug 2022 – Jul 2023

- Redesigned and standardized the **production capacity planning tool** of a global pharmaceutical manufacturer through an automated, KPI-oriented model and rolled it out in **6 plants (13 production lines)**.
- Optimized production batch sizes across **9 plants**, saving **\$800k/year (1.1%** of conversion cost).
- Optimized the production plan and timelines of a train manufacturer, recovering **2.5 months** of delivery delays worth **\$1M** in saved penalties.

Sr Program Manager, 3PL Ops Excellence & Quality | Amazon (Luxembourg)

Sept 2017 – Aug 2022

- Led the planning and execution of multi-item order shipping capabilities across **10 3PL sites**, generating **\$10Mn+** in transport savings.
- Designed network-wide **inventory audit programs** across **25+ warehouses**, improving the Inventory Record Defect Rate metric by **50%**.
- Recovered a severe inventory defects crisis from **120k** to **<10k DPMO** in **8 weeks**, driving end-to-end process redesign.
- Led root cause investigation and settlement negotiation on a **\$7M** disputed inventory loss, protecting **3PL** partner relationships.

EDUCATION

- **MSc in Aerospace Engineering** | Universidad Carlos III de Madrid & TU Delft (Exchange)
- **BSc in Aerospace Engineering & BSc in Mechanical Engineering** | Universidad de León
- **[Certificate of Proficiency: ROS2 Basics \(Python\)](#)** | The Construct Robotics Institute (2026)

LANGUAGES

- Spanish — Native | English — Proficient (C1) | French — Intermediate (B2)